CRISPR-Cas9 in Human Health

Where the Science Actually Stands — Therapeutic Progress, Safety Trade-offs, and the Ethics That Come With Them

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ABSTRACT

CRISPR Cas9 moved from laboratory based genome editing technique to clinical treatment several genetic disorders Current clinical evidence is strongest for blood disorders treated through ex vivo editing and liver diseases treated through in vivo approaches Exagamglogene autotemcel for example has shown sustained benefits in patients with sickle cell disease and β thalassemiaLiver-directed CRISPR therapies have also produced promising results in hereditary transthyretin amyloidosis and hereditary angioedema.Other applications including inherited eye diseases cancer immunotherapy and cardiometabolic disorders are still being investigated. Key challenges include unintended genetic changes efficient delivery to target cells immune responses and limited long term safety dataEthical concerns also remain important particularly the distinction between somatic and embryo editing informed consent for irreversible treatments equitable access to expensive gene therapies progress will depend on improving editing accuracy and delivery monitoring patients over the long term and developing effective regulatory frameworks that can keep pace with the technology

Keywords: *CRISPR Cas9 genome editing precision medicine clinical applications gene therapy ethics*

1. INTRODUCTION

Genome editing predates CRISPR Cas9 with earlier technologies such as zinc finger nucleases and transcription activator-like effector nucleases (TALENs) providing the foundation for targeted genetic modification However these methods were relatively complex costly and time consuming to design CRISPR-Cas9 simplified genome editing by using a guide RNA to direct the Cas9 enzyme to a specific DNA sequence making the system more programmable and accessible This flexibility has accelerated its development for therapeutic applications, including inherited blood disorders metabolic diseases eye disorders cancer and infectious diseases

Despite its potential several challenges remain before CRISPR Cas9 can be widely adopted in clinical practice Longterm safety unintended genetic changes informed consent and equitable access to treatment continue to require careful consideration This review examines the current therapeutic applications of CRISPR Cas9 the technical and safety limitations affecting its wider use and the ethical and regulatory considerations associated with its clinical application

Objectives

- Examine the therapeutic applications of CRISPR Cas9 across major disease categories
- Identify the key technical and safety challenges limiting broader clinical use

2. METHODS

This scoping review was conducted using the Arksey and O Malley framework and reported in accordance with PRISMA ScR guidelines PubMed Embase and Dimensions were searched for English language peer reviewed studies A total of 3,013 records were identified during the initial search. Following the screening and selection process 32 studies were included in the review These comprised preclinical studies Phase I–III clinical trials and literature addressing ethical and regulatory issues The selected studies were thematically synthesized across three main areas clinical applications safety limitations and governance This approach allowed the review to map the breadth of existing evidence while identifying key gaps for future research

3. THERAPEUTIC APPLICATIONS

3.1 Hematologic and Monogenic Disorders

Blood disorders represent one of the most advanced clinical applications of CRISPR Cas9 with exagamglogene autotemcel serving as a prominent example Rather than directly correcting the mutated HBB gene exa cel targets an enhancer involved in the regulation of BCL11A in a patient's hematopoietic stem cells Disrupting this enhancer increases fetal hemoglobin production which reduces the tendency of sickle hemoglobin to polymerize The editing process is performed ex vivo allowing the modified cells to be assessed before they are returned to the patient This approach also provides greater control over the edited cell population and allows potential abnormalities to be identified before infusion

Figure 1 walks through that process from stem-cell collection to long-term follow-up.

There's momentum beyond exa cel too correcting βthalassemia mutations directly in patient-derived iPSCs reactivating fetal hemoglobin via SOX6 and early correction work for severe congenital neutropenia and Artemis deficient immunodeficiency Taken together these point to ex vivo stem cell editing becoming a general strategy across inherited blood and immune disorders not just a fix for one disease.

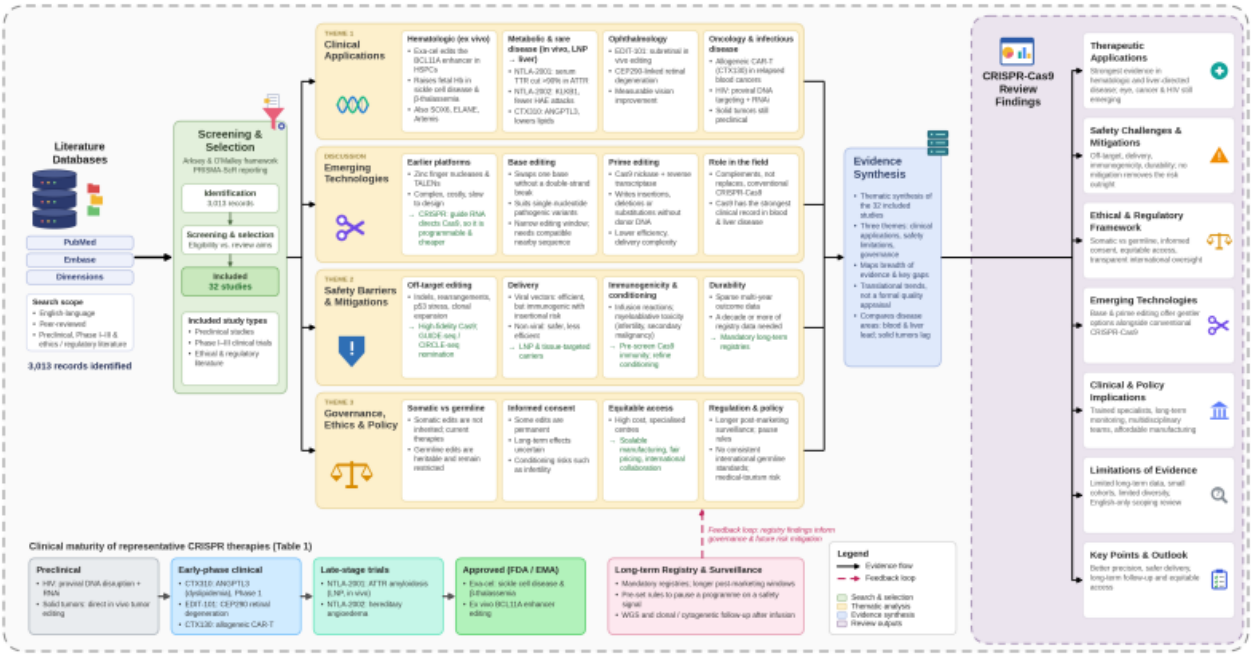


Figure 1. CRISPR-Cas9 review architecture showing literature screening and selection, thematic synthesis of clinical applications, safety barriers, and governance, and the resulting review findings.

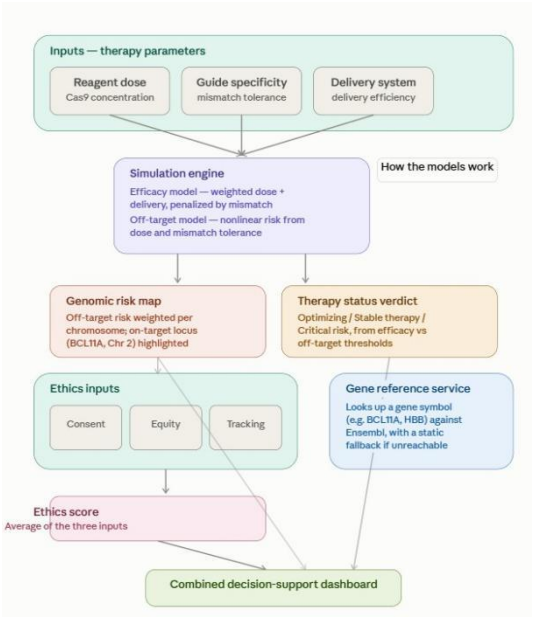


Figure 2. Simulation model workflow: therapy parameters and ethics inputs feed the simulation engine, genomic risk map and therapy-status verdict, combined in a decision-support dashboard.

3.2 In Vivo Editing for Metabolic and Rare Diseases

Delivering CRISPR-Cas9 directly into the body — mostly via lipid nanoparticles aimed at the liver — has opened up therapy beyond cell-based approaches. NTLA-2001 and related agents cut serum transthyretin by more than 90 percent after a single infusion in patients with hereditary transthyretin amyloidosis, and the effect held up over time. NTLA-2002 targets KLKB1 and reduced attack frequency in hereditary angioedema. CTX310, going after ANGPTL3, has shown early promise lowering lipids for cardiovascular risk. Liver-directed in vivo editing is arguably the most scalable near-term model for preventing chronic disease at a population level, simply because the liver is easy to reach and lipid nanoparticles work well there.

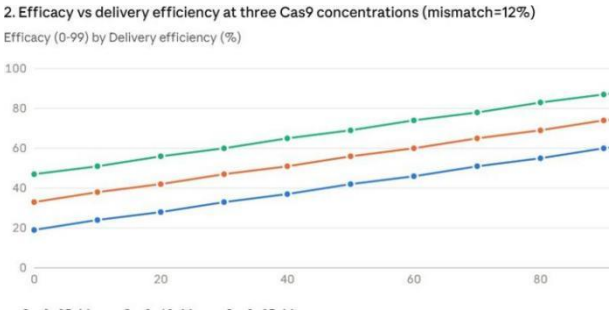


Figure 3. Efficacy vs delivery efficiency at three Cas9 concentrations.

3.3 Ophthalmology, Oncology, and Infectious Disease

EDIT101 was one of the first real demonstrations that in vivo editing works in the eye producing measurable vision improvement in CEP290 linked retinal degeneration The use of CRISPR Cas9 in cancer treatment remains challenging particularly for solid tumors because of difficulties in targeted delivery and the complex nature of tumor tissues Greater progress has been seen in cell based cancer therapies where genetically modified allogeneic CAR T cells with multiple receptor modifications have shown promising early results in patients with relapsed blood cancers

CRISPR based approaches are also being investigated for infectious diseases In HIV research targeting proviral DNA using CRISPR has shown potential antiviral effects particularly when combined with RNA interference However viral escape and resistance remain important challenges Table 1 summarizes the current development of these applications

Table 1. Representative CRISPR Cas9 therapies by disease category

Disease	Representative Therapy / Approach	Clinical Stage
Sickle cell disease / β-thalassemia	Exa-cel — ex vivo BCL11A enhancer editing	Approved (FDA/EMA)
ATTR amyloidosis	NTLA-2001 — in vivo TTR knockdown (LNP delivery)	Late-stage / approved indications
Hereditary angioedema	NTLA-2002 — in vivo KLKB1 editing	Late-stage trials
Dyslipidemia / cardiovascular risk	CTX310 — in vivo ANGPTL3 editing	Early-phase (Phase 1)
Retinal degeneration (CEP290)	EDIT-101 — subretinal in vivo editing	Early-phase clinical
Hematologic malignancy	Allogeneic CRISPR CAR-T (e.g., CTX130)	Early-phase clinical
HIV / infectious disease	Proviral DNA disruption + RNAi	Preclinical
Solid tumors	Direct in vivo tumor editing	Preclinical

4. SAFETY CHALLENGES AND TECHNICAL BARRIERS

Off target editing remains one of the major safety concerns associated with CRISPR Cas9 Since many current approaches involve cutting both strands of DNA unintended effects can extend beyond simple changes at incorrect sites These may include larger genomic rearrangements at similar DNA sequences activation of p53 related cellular stress responses and in some cases the expansion of specific edited blood cell clones Such risks highlight the need for improved editing accuracy and long term monitoring of treated patients

High-fidelity Cas9 variants, off-target nomination assays, and ongoing clonal surveillance all help, but none of them close the risk out completely. Delivery is the second recurring bottleneck: viral vectors transduce cells efficiently but carry immunogenicity and insertional risk, while non-viral systems are safer but less efficient. On top of that, the myeloablative conditioning many ex vivo therapies require carries its own risks — infertility and secondary malignancy among them. And because most of these therapies are still new, the field genuinely needs several more years — ideally a decade or more — of registry data before anyone can speak with the same confidence about durability that they can already speak with about short-term efficacy.

Regulators have responded by requiring longer post-marketing surveillance windows, mandatory long-term registries, and pre-set rules for pausing a program if a safety signal shows up. Table 2 lines up the main risk categories against how each is currently being managed. Worth noting: none of these mitigations eliminates its corresponding risk outright, which is a big part of why safety — not efficacy — is now the real bottleneck slowing broader rollout.

Table 2. Principal safety barriers and current mitigations.

Barrier	Concern	Current Mitigation
Off-target editing	Indels, structural rearrangements, oncogenic risk	High-fidelity Cas9 variants; GUIDE-seq / CIRCLE-seq off-target nomination
Delivery	Viral vector immunogenicity vs. non-viral inefficiency	Lipid nanoparticle (LNP) and tissue-targeted carrier development
Immunogenicity	Infusion reactions; conditioning-regimen toxicity	Pre-screening for Cas9 immunity; refined conditioning regimens
Durability	Sparse multi-year outcome data	Mandatory registry-based long-term surveillance

The safety concerns that come up again and again in the literature:

- Off-target cleavage and structural genomic rearrangements
- Immune responses to Cas9 protein and delivery vehicles
- Toxicity from myeloablative conditioning regimens
- Still-incomplete multi-year safety surveillance

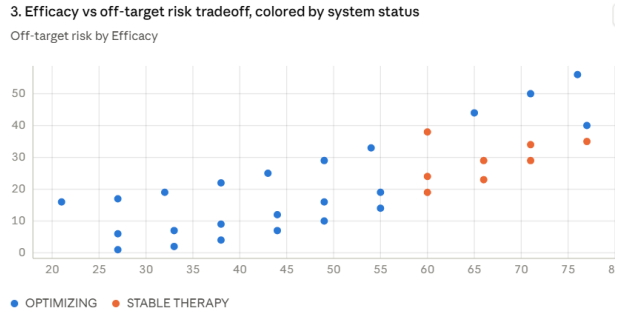


Figure 4. Efficacy vs off-target risk tradeoff, colored by system status.

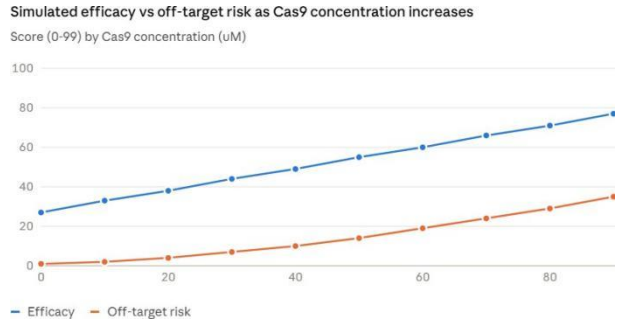


Figure 5. Simulated efficacy vs off-target risk as Cas9 concentration increases.

5. ETHICAL, SOCIETAL, AND REGULATORY CONSIDERATIONS

One of the main ethical issues in genome editing is the difference between somatic and germline editing Somatic changes affect only the treated patient and are not inherited whereas germline changes can be passed on to future generations Because of these potential long term effects heritable genome editing continues to raise significant ethical and scientific concerns Current CRISPR Cas9 therapies in clinical development mainly involve somatic editing where the potential risks can be assessed through clinical studies and patient monitoring

Some ex vivo treatments result in permanent genetic changes after the edited cells are introduced into the patient

This makes informed consent particularly important as patients need to understand that some long term effects may remain uncertain The conditioning procedures used in certain treatments can also carry serious risks including infertility

Access to these therapies is another concern Diseases such as sickle cell disease affect large populations while CRISPR based treatments can be expensive and require specialized healthcare facilities Without efforts to improve affordability and availability advances in genome editing may not benefit patients equally

Long term follow up transparent reporting and consistent regulatory standards are therefore important for the responsible use of CRISPR Cas9 Key ethical considerations include maintaining a clear distinction between somatic and germline editing ensuring informed consent reflects uncertainty about long term outcomes improving equitable access and strengthening transparent international oversight

6. DISCUSSION

The clinical development of CRISPR Cas9 differs considerably across disease areas Hematologic disorders have progressed more rapidly because they offer several practical advantages including well defined genetic targets accessible stem cells established transplantation methods and measurable clinical outcomes such as transfusion independence Liver directed therapies have also advanced because the liver can be readily targeted using delivery systems such as lipid nanoparticles

In contrast, the development of CRISPR based treatments for solid tumors remains more challenging. Tumor heterogeneity, an immunosuppressive environment, and difficulties in delivering the editing system to the appropriate cells have limited progress, with much of the research still focused on preclinical studies and cell-based therapies.

Compared with earlier genome-editing technologies such as zinc finger nucleases and TALENs, CRISPR Cas9 offers greater programmability and relatively lower development costs. However, its wider accessibility also increases the need for appropriate ethical and regulatory oversight. This review focuses on major translational trends rather than providing a formal assessment of the quality of the individual studies included.This review is a map of translational trends, not a formal quality appraisal of the underlying studies, and it reflects the English-language literature, which likely underrepresents work happening in other regions.

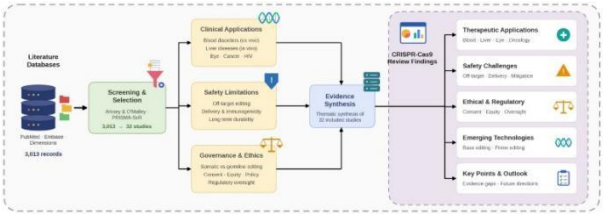


Figure 6. Molecular mechanism, manufacturing workflow, and safety surveillance across the CRISPR-Cas9 therapeutic lifecycle. The feedback loop connects long-term registry findings with governance and future risk mitigation.

7. EMERGING EDITING TECHNOLOGIES

Conventional CRISPR-Cas9 is still the most clinically advanced platform, but its reliance on cutting both DNA strands is exactly what's driving interest in gentler alternatives. Base editing swaps one DNA base for another without a double-strand break, which suits it well to fixing the single-nucleotide changes behind most pathogenic variants — though it only works within a fairly narrow editing window and needs compatible target sequences nearby. Prime editing goes a step further, pairing a Cas9 nickase with a reverse transcriptase to write insertions, deletions, or substitutions without needing donor DNA at all — at the cost of lower efficiency in many cell types and more delivery complexity. Neither is a replacement for standard CRISPR-Cas9 so much as a complement to it; conventional CRISPR still carries the strongest clinical track record in hematologic and liver-directed disease.

8. CLINICAL AND POLICY IMPLICATIONS

As genome editing moves out of specialized trial centers and into routine care how well health systems are prepared will Responsible implementation of CRISPR based therapies requires more than technological advances Healthcare systems need trained specialists long term patient monitoring and multidisciplinary teams to manage treatment and follow up care A major policy challenge is the lack of consistent international standards for germline editing and the commercialization of genome editing therapies Differences between countries may create regulatory gaps and increase the risk of unethical practices including unregulated medical tourism

The high cost of individualized manufacturing and conditioning procedures can also limit access particularly for patients in lower resource settings Improving affordability will require scalable manufacturing methods appropriate pricing strategies and stronger international collaboration Ensuring equitable access should therefore remain an important part of the development and regulation of CRISPR based therapies

9. LIMITATIONS OF CURRENT EVIDENCE

This review has several limitations Long term safety data remain limited because many CRISPR Cas9 therapies have only recently entered clinical trials Some studies also involve small patient groups which may limit the generalizability of their findings In addition the geographic and demographic diversity of trial participants is still limited making it difficult to determine whether the reported outcomes apply equally across different populations

Applications in oncology infectious diseases and neurological disorders are also largely at the preclinical stage highlighting the uneven progress of CRISPR Cas9 across disease areas As this is a scoping review its purpose is to describe the available evidence rather than formally assess study quality or directly compare treatment effectiveness The findings should therefore be viewed as an overview of the current field rather than a definitive assessment of individual therapies Finally the restriction to English language publications may have excluded relevant

research from other regions Future reviews should consider including studies published in multiple languages to provide a broader perspective

10. CONCLUSION

CRISPR Cas9 has progressed from an experimental laboratory technique to a clinically validated approach The strongest clinical evidence currently comes from selected monogenic blood disorders and liver targeted diseases Expanding its application to cancer infectious diseases and other conditions will require improved editing precision safer and more efficient delivery systems and longer term monitoring of treatment outcomes Ethical and regulatory frameworks must also develop alongside these advances

The future of CRISPR based medicine will depend on balancing scientific progress with responsible clinical use and equitable access Addressing safety regulation and accessibility together will be essential for the wider and sustainable use of genome-editing therapies

KEY POINTS

- Hematologic and liver targeted monogenic disorders currently have the strongest clinical evidence
- Off target effects and delivery challenges remain major technical concerns
- Somatic genome editing therapies are advancing clinically while heritable germline editing remains restricted
- Equitable access should be considered throughout the development of CRISPR based therapies

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Ethics Statement. Ethical approval was not required because the review used previously published research and did not involve human participants patient data or animal experimentation

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